Computer Networks

Homework-2

Koneti Karthik

230568

1. **Within the IP packet header, what is the value in the upper layer protocol field?**

   Answer: The value in the upper layer protocol field is **1 (ICMP)**.

2. **How many bytes are in the IP header? How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.**

   Answer: The IP header length is **20 bytes**. The total length field shows **56 bytes**.

   Payload size = Total Length – Header Length

   $$56 - 20 = 36 bytes$$

   Therefore, the payload size is **36 bytes**.

3. **Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.**

   Answer: No, the datagram has not been fragmented. The **More Fragments (MF)** flag is not set and the **Fragment Offset** is 0, indicating that this is a complete datagram.

4. **What is the value in the Identification field and the TTL field?**

   Answer: Identification field = **0x0b02 (3296)** TTL field = **1**

5. **Can you say whether the message corresponding to the above packet has been fragmented?**

   Answer: Yes, the message has been fragmented because the **More Fragments (MF)** flag is set.

6. **What information in the IP header indicates that the datagram has been fragmented?**

   Answer: The datagram is fragmented because:

   - The **More Fragments (MF)** flag is set.
   - The **Fragment Offset** field may be non-zero.

7. **What information in the IP header indicates whether this is the first fragment versus a latter fragment?**

   Answer: If the **Fragment Offset** is 0, it is the first fragment. If the Fragment Offset is greater than 0, it is a latter fragment.

8. **What information in the IP header indicates that this is not the first datagram fragment?**

   Answer: The **Fragment Offset** field is greater than 0, indicating that it is not the first fragment.

9. **Are there more fragments? How can you tell?**

   Answer: No, there are no more fragments because the **More Fragments (MF)** flag is not set.

10. **If Fig. 2 and Fig. 3 are the 1st and 2nd fragments of a message, then what fields change in the IP header between the first and second fragment?**

    Answer: The fields that change between fragments are:

    - **Fragment Offset**
    - **More Fragments (MF) flag**
    - **Total Length**
    - **Header Checksum**